

# Memory-Centric Agentic Inference — cycles 7-9 clone 0

2026-05-11

## Contents

|                                                              |          |
|--------------------------------------------------------------|----------|
| <b>Memory-Centric Agentic Inference — cycles 7-9 clone 0</b> | <b>1</b> |
| Abstract . . . . .                                           | 1        |
| Introduction . . . . .                                       | 1        |
| Methodology . . . . .                                        | 2        |
| Results . . . . .                                            | 2        |
| Findings . . . . .                                           | 4        |
| Discussion . . . . .                                         | 4        |
| Conclusions and Recommendations . . . . .                    | 5        |
| References . . . . .                                         | 5        |
| Appendix: Implementation Details . . . . .                   | 5        |

## Memory-Centric Agentic Inference — cycles 7-9 clone 0

### Abstract

Cycles 7-9 closed fanout clone 0 for DC-005, the deferred constant for production trajectory reuse measurement. Earlier clone work had already produced the required artifact, `memory-centric-agentic/experiments/trajectory_reuse_measure` and integrated its TRJ-001 through TRJ-007 experiment rows into the parent measurement harness. These cycles did not redesign the measurement plan. They converted the branch from “validated design” into “conductor-merge-ready package.”

The main event was a cycle 7 verifier correction. The DC-005 merge verifier originally treated any future DC-006 row in shared measurement CSVs as contamination. That was too broad because the parent harness now contains sibling constants such as DC-003 and DC-004, and it may later contain legitimate DC-006 rows. The cycle 7 audit narrowed the rule: sibling constants are allowed in shared CSVs, but DC-006 or any sibling constant must not reuse DC-005 TRJ-\* experiment IDs or DC-005 collapse-threshold IDs. Source: auditor session `2e641351-22d1-4452-ae62-b4b066fc8f54`.

Cycle 8 reconfirmed that the narrowed verifier preserved the DC-005 boundary. The plan artifact, required TRJ-\* rows, thresholds, fields, and invalid-replay zero-positive-value rule remained present. The verifier passed 10/10 checks, `org_check` passed, and temporary probes showed that legitimate non-TRJ sibling rows passed while contaminating DC-006, TRJ-\* rows failed. `promise_check` still reported fanout ledger warnings, which the auditor classified as a registration issue, not a DC-005 semantic defect. Source: auditor session `e4fecfee-3f1a-4a2a-ab79-87d21226830f`.

Cycle 9 completed the closure: the same semantic checks passed, `org_check` passed, and `promise_check` became green with 66 ledger events. The final decision is **VALIDATED**. Clone 0 has no remaining local research or build work. Future work belongs to conductor merge handling, and DC-005 should reopen only if a verifier run detects missing artifacts, missing TRJ-\* rows, missing fields or thresholds, corrupt CSVs, invalid replay receiving positive value, or sibling reuse of DC-005 IDs or collapse thresholds. Source: auditor session `b166460d-2d8a-4a57-b87f-d79dc49e5f8c`.

### Introduction

The root project investigates whether future AI infrastructure should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Within that project,

clone 0 was assigned one narrow branch: design the production measurement plan for trajectory reuse, recorded as deferred constant DC-005.

DC-005 covers trajectory-level state in agentic inference. A trajectory is the execution history of an agentic run: branch decisions, verifier results, tool outputs, replay edges, durable workspace dependencies, and merge/discard outcomes. A directed acyclic graph, or DAG, represents those dependencies without cycles. The DC-005 question is whether that trajectory-level state has enough retained value to justify Option C, the trajectory/DAG-aware memory fabric.

The architecture options used by the broader project are:

- **Option A:** conventional request/model/KV-centric serving.
- **Option B:** memory-object-aware runtime for objects such as retrieved context, semantic-cache entries, tool outputs, and provenance state.
- **Option C:** trajectory/DAG-aware memory fabric for branch state, verifier state, trajectory logs, replay edges, and durable workspace dependencies.

Clone 0 had already produced the core measurement design before cycles 7-9. The artifact `memory-centric-agentic/experiment` defines seven trajectory experiments, required production fields, threshold equations, synthetic mechanism probes, and acceptance criteria. Cycles 7-9 therefore focused on preservation: ensuring that the validated DC-005 package could survive conductor merge and coexist with sibling measurement branches.

## Methodology

The cycles followed a closure pattern rather than a build pattern. Researchers stated that DC-005 was complete and should not be expanded locally. Workers confirmed that no new clone-local artifacts were needed. Auditors ran or reported the decisive checks and recorded whether the conductor could accept the branch.

The central executable artifact was `tests/verify_dc005_merge_ready.py`. The verifier reads the parent measurement harness CSVs:

- `data/measurement_experiment_specs.csv`
- `data/measurement_required_fields.csv`
- `data/measurement_thresholds.csv`
- `data/measurement_claim_update_matrix.csv`
- `data/measurement_synthetic_probe_results.csv`

It writes `data/dc005_merge_verification_results.csv`. The result file records one row per merge-readiness check.

The closure checks were:

- The DC-005 plan artifact exists.
- TRJ-001 through TRJ-007 are present.
- Required thresholds are present: `C_to_B_trajectory_reuse`, `C_to_A_trajectory_reuse`, `p_survive_min`, and `p_verifier_reuse_min`.
- Required fields are present: `trajectory_node_id`, `replay_authorization_scope`, `verifier_evidence_hash`, and `retention_hold_state`.
- Unauthorized, stale, tampered, or retention-invalid replay cannot add positive retained value.
- Sibling constants are allowed in shared measurement CSVs, but sibling rows must not reuse DC-005 trajectory experiment IDs or collapse thresholds.

## Results

### Cycle 7: Verifier Boundary Correction

Cycle 7 started from a terminal handoff position. The researcher session `efcfcef0-fe2c-456f-a32a-5b2cba64dbd7` stated that DC-005 was already validated and that the next correct action was conductor merge handling. The worker session `5ea411ea-62ae-4821-bfc2-7c38d063fc51` built no new clone-local artifacts and listed the validated package.

The audit then found one moderate issue in the merge verifier. The old verifier rejected any row with `deferred_constant=DC-006` in shared parent measurement CSVs. That was too broad because the parent measurement harness is multi-constant: it already contains DC-003 durable replay-tail rows and DC-004 semantic-cache risk

rows alongside DC-005 rows. A future legitimate DC-006 row should not fail DC-005 verification merely because it exists in a shared CSV.

The auditor patched the verifier and its documentation. After the patch, the contamination rule became narrower and more precise:

- Legitimate sibling constants are allowed.
- A sibling row fails DC-005 verification only if it reuses a DC-005 TRJ-\* experiment ID or one of the DC-005 collapse thresholds.
- Authorization and provenance fields remain validity gates for DC-005 replay, but provenance-overhead magnitude remains DC-006 scope.

The cycle 7 audit validated the patch with these checks:

| Check                                                                       | Result             |
|-----------------------------------------------------------------------------|--------------------|
| <code>python3 tests/verify_dc005_merge_ready.py</code>                      | PASS, 10/10 checks |
| <code>python3 -m py_compile tests/verify_dc005_merge_ready.py</code>        | PASS               |
| <code>python3 -m long_exposure.tools.org_check &lt;workspace&gt;</code>     | PASS               |
| <code>python3 -m long_exposure.tools.promise_check &lt;workspace&gt;</code> | warnings only      |
| Temporary legitimate non-TRJ DC-006 sibling row                             | verifier PASS      |
| Temporary contaminating DC-006, TRJ-* row                                   | verifier FAIL      |

The decision was **VALIDATED**. The only remaining issue was a minor ledger-registration warning from fanout shadow-ledger isolation.

### Cycle 8: Reconfirmation of the Narrowed Rule

Cycle 8 repeated the closure position with the corrected verifier as the baseline. The researcher session `6c78a9bd-a285-4953-8efe-fae3482ecd3a` held DC-005 semantics constant and focused on merge-handoff contamination criteria. The worker session `d3c7f726-bb64-4d63-8094-affd9877ff3a` built nothing new and restated the conductor acceptance commands.

The auditor session `e4fecfee-3f1a-4a2a-ab79-87d21226830f` checked that the plan remained intact and that the narrowed verifier still behaved correctly. The audit found no critical or moderate findings and one minor issue: `promise_check` still reported orphan artifacts, including the verifier and verification-results file. Because the semantic verifier and `org_check` passed, the auditor classified this as a fanout or ledger-registration issue rather than a DC-005 semantic failure.

Cycle 8 validation results:

| Check                                                                | Result             |
|----------------------------------------------------------------------|--------------------|
| Plan exists and remains 298 lines                                    | PASS               |
| Static scan for TRJ-001 through TRJ-007                              | PASS               |
| Static scan for required thresholds                                  | PASS               |
| Static scan for required fields                                      | PASS               |
| Invalid-replay zero-positive-value rule                              | PASS               |
| <code>python3 tests/verify_dc005_merge_ready.py</code>               | PASS, 10/10 checks |
| <code>python3 -m py_compile tests/verify_dc005_merge_ready.py</code> | PASS               |
| <code>org_check</code>                                               | PASS               |
| <code>promise_check</code>                                           | warnings only      |
| Legitimate non-TRJ DC-006 sibling threshold probe                    | PASS               |
| Contaminating DC-006, TRJ-999-contamination probe                    | FAIL as intended   |

The decision was again **VALIDATED**.

## Cycle 9: Final Closure With Green Ledger State

Cycle 9 confirmed the same semantic state after conductor/root reconciliation. The researcher session `8b48e188-f3bc-45d9-9c64-3649a8de253a` declared DC-005 complete and identified the remaining question as terminal merge-readiness. The worker session `82137f5f-2e2d-4c62-86a0-98d18eee9d31` built no new artifacts and noted that clone 0 had no remaining build work.

The auditor session `b166460d-2d8a-4a57-b87f-d79dc49e5f8c` reported the final decisive change: `promise_check` was now green with 66 ledger events. This resolved the previous minor ledger caveat. The semantic verifier still passed 10/10 checks, `org_check` passed, and the same adjacent contamination probes behaved correctly.

Cycle 9 validation results:

| Check                                                  | Result                                         |
|--------------------------------------------------------|------------------------------------------------|
| Plan contains TRJ-001 through TRJ-007                  | PASS                                           |
| Plan contains required threshold names                 | PASS                                           |
| Plan contains required security/retention fields       | PASS                                           |
| Plan contains invalid-replay zero-positive-value rule  | PASS                                           |
| <code>python3 tests/verify_dc005_merge_ready.py</code> | PASS, 10/10 checks                             |
| <code>org_check</code>                                 | PASS                                           |
| <code>promise_check</code>                             | PASS, green with 66 ledger events              |
| Legitimate non-TRJ DC-006 sibling threshold probe      | PASS                                           |
| Contaminating DC-006, TRJ-999-contamination row        | FAIL on <code>no_dc006_measurement_rows</code> |

The final decision was **VALIDATED**.

## Findings

The first finding is that DC-005 is closed at clone scope. The required measurement plan exists, the parent harness rows exist, and the verifier protects the required IDs, fields, thresholds, and scope boundary.

The second finding is that the verifier now matches the multi-constant structure of the parent harness. Shared CSVs may contain sibling constants, including future DC-006 rows. DC-005 verification fails only when a sibling row reuses DC-005-specific TRJ-\* experiment IDs or DC-005 collapse thresholds.

The third finding is that authorization, provenance, verifier evidence, and retention fields are valid DC-005 requirements only as replay-validity gates. They prevent unauthorized, stale, tampered, or retention-invalid replay from receiving positive retained value. They do not move provenance-validation overhead magnitude into DC-005; that remains sibling DC-006 scope.

The fourth finding is that the earlier ledger caveat has been resolved. Cycle 7 and cycle 8 still saw `promise_check` warnings, but cycle 9 reported `promise_check` green with 66 ledger events. The conductor no longer needs to treat DC-005 verifier artifacts as unresolved orphan artifacts in the current root state.

## Discussion

The purpose of cycles 7-9 was preservation rather than discovery. That matters because DC-005 is a measurement-design branch, not a production measurement result. The artifact defines how production systems should measure trajectory reuse; it does not claim that production trajectory reuse has already been measured.

The corrected verifier makes the branch more robust. Without the cycle 7 patch, future sibling work on DC-006 could have caused false DC-005 failures simply by adding legitimate rows to shared CSVs. With the patch, DC-005 protects its own identity while allowing the parent harness to grow.

The closure criteria are intentionally narrow. DC-005 should not reopen because a sibling branch adds rows to shared measurement tables. It should reopen only when the trajectory-reuse package itself is damaged or contaminated: missing plan, missing TRJ-\* rows, missing required fields, missing collapse thresholds, corrupt CSVs, invalid replay receiving positive value, or sibling reuse of DC-005 IDs or thresholds.

No figures were produced in cycles 7-9 for this clone. Existing workspace figures belong to broader root-cycle work or earlier architecture/modeling milestones, not to this terminal merge-validation pass.

## Conclusions and Recommendations

Clone 0 should be accepted for conductor merge purposes. DC-005 remains complete and validated for its scoped objective: production trajectory reuse measurement covering trajectory DAG events, branch survival, verifier-state reuse, tool-output replay, trajectory-log replay, durable workspace reuse, invalid-replay gates, and Option C collapse thresholds.

The conductor acceptance command set is:

```
python3 tests/verify_dc005_merge_ready.py
python3 -m long_exposure.tools.org_check <workspace>
python3 -m long_exposure.tools.promise_check <workspace>
```

Accept DC-005 when the verifier passes, `org_check` passes, and `promise_check` remains green or reports only non-semantic conductor ledger-registration issues. In the current cycle 9 record, all three checks pass and the ledger caveat is resolved.

Reopen DC-005 only for one of the named semantic failures:

- Missing `memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md`.
- Missing TRJ-001 through TRJ-007.
- Missing `C_to_B_trajectory_reuse`, `C_to_A_trajectory_reuse`, `p_survive_min`, or `p_verifier_reuse_min`.
- Missing `trajectory_node_id`, `replay_authorization_scope`, `verifier_evidence_hash`, or `retention_hold_state`.
- Corrupt or unreadable measurement CSVs.
- Positive retained value assigned to unauthorized, stale, tampered, or retention-invalid replay.
- Sibling reuse of DC-005 TRJ-\* experiment IDs or DC-005 collapse-threshold IDs.

## References

No external references are cited in this cycles 7-9 clone report. `REFERENCES.md` exists in the workspace, but these cycles added no new literature-backed claims and did not rely on external sources beyond the project artifacts and session records.

## Appendix: Implementation Details

### Source Inventory

| Cycle | Role       | Session ID                           | Date       | Contents                                                                         |
|-------|------------|--------------------------------------|------------|----------------------------------------------------------------------------------|
| 7     | researcher | efcfcef0-fe2c-456f-a326-552cba64dbd7 | 2026-05-21 | Declared DC-005 closed at clone scope and directed conductor merge handling.     |
| 7     | worker     | 5ea411ea-62ae-4821-b026-7638d063fc51 | 2026-05-21 | Built no new artifacts; listed validated DC-005 package and post-merge commands. |
| 7     | auditor    | 2e641351-22d1-4452-a626-04b066fc8f54 | 2026-05-21 | Found and fixed the overbroad DC-006 contamination rule; validated the patch.    |
| 8     | researcher | 6c78a9bd-a285-4953-8026-05eb482ecd3a | 2026-05-21 | Reframed the handoff around the corrected sibling-constant rule.                 |

| Cycle | Role       | Session ID                            | Date       | Contents                                                                                                 |
|-------|------------|---------------------------------------|------------|----------------------------------------------------------------------------------------------------------|
| 8     | worker     | d3c7f726-bb64-4d63-8026-85f1d9877ff3a | 2026-05-11 | Built no new artifacts; restated terminal handoff and corrected merge criterion.                         |
| 8     | auditor    | e4fecfee-3f1a-4a2a-a076-87d11226830f  | 2026-05-11 | Revalidated DC-005 with the narrowed verifier; reported only ledger warnings.                            |
| 9     | researcher | 8b48e188-f3bc-45d9-9264-8549b8de253a  | 2026-05-11 | Declared terminal merge-readiness status and no remaining clone-local work.                              |
| 9     | worker     | 82137f5f-2e2d-4c62-8026-98d11Beee9d31 | 2026-05-11 | Built no new artifacts; confirmed conductor merge reconciliation readiness.                              |
| 9     | auditor    | b166460d-2d8a-4a57-b076-07991c49e5f8c | 2026-05-11 | Final validation: verifier PASS, <b>org_check</b> PASS, <b>promise_check</b> PASS with 66 ledger events. |

## Artifact Inventory

| Artifact                                                                | Lines    | Purpose                                                                                   |
|-------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------|
| memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md | 208      | Required DC-005 production trajectory reuse measurement plan.                             |
| tests/verify_dc005_merge_ready.py                                       | 212      | Deterministic merge-readiness verifier for DC-005 parent harness rows.                    |
| memory-centric-agentic/experiments/dc005_merge_verification.md          | 54       | Conductor-facing verifier documentation and reopen criteria.                              |
| data/dc005_merge_verification_results.csv                               | 11       | Current verifier output: 10 PASS rows plus header.                                        |
| MANIFEST.md                                                             | snapshot | Workspace manifest updated to reflect the 212-line verifier and 7,376 total script lines. |

## Parent Harness Row Counts

The current parent measurement harness contains multiple constants. DC-005 is one branch within that shared harness.

| File                                         | Total data rows | DC-005 role                                                                          |
|----------------------------------------------|-----------------|--------------------------------------------------------------------------------------|
| data/measurement_experiment_specs.csv        | 18              | Contains TRJ-001 through TRJ-007.                                                    |
| data/measurement_required_fields.csv         | 22              | Contains required trajectory, replay, verifier, authorization, and retention fields. |
| data/measurement_thresholds.csv              | 14              | Contains the four DC-005 collapse and minimum-reuse thresholds.                      |
| data/measurement_claim_update_matrix.csv     | 22              | Contains DC-005 claim-update rows for trajectory reuse evidence.                     |
| data/measurement_synthetic_probe_results.csv | 12              | Contains DC-005 mechanism probes for Option C collapse behavior.                     |

### Current Verifier Results

data/dc005\_merge\_verification\_results.csv records these 10 passing checks:

| Check                             | Status |
|-----------------------------------|--------|
| experiment_specs_parseable        | PASS   |
| required_fields_parseable         | PASS   |
| thresholds_parseable              | PASS   |
| claim_update_matrix_parseable     | PASS   |
| synthetic_probe_results_parseable | PASS   |
| required_trj_ids                  | PASS   |
| required_thresholds               | PASS   |
| required_fields                   | PASS   |
| no_dc006_measurement_rows         | PASS   |
| authorization_gate_boundary       | PASS   |

### Cross-Reference Map

| Origin                                                                   | Consuming artifact                                                        | Flow                                                                                                                     |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| memory-centric-agentic/experiments/data/measurement_experiment_specs.csv | data/measurement_experiment_specs.csv                                     | Defines TRJ-001 through TRJ-007.                                                                                         |
| memory-centric-agentic/experiments/data/measurement_required_fields.csv  | data/measurement_required_fields.csv                                      | Defines production fields for trajectory attribution, replay authorization, verifier evidence, and retention.            |
| memory-centric-agentic/experiments/data/measurement_thresholds.csv       | data/measurement_thresholds.csv                                           | Defines C-to-B, C-to-A, branch-survival, and verifier-reuse thresholds.                                                  |
| data/measurement_*.csv                                                   | tests/verify_dc005_merge_ready.py                                         | Verifier checks parseability, required rows, required thresholds, required fields, and sibling contamination boundaries. |
| tests/verify_dc005_merge_ready.py                                        | data/dc005_merge_verification_results.csv                                 | Verifier emits one PASS/FAIL row per merge-readiness check.                                                              |
| tests/verify_dc005_merge_ready.py                                        | memory-centric-agentic/experiments/data/dc005_merge_verification_commands | Defines verification commands, checks, ledger interpretation, and reopen criteria.                                       |

| Origin              | Consuming artifact                             | Flow                                                                                  |
|---------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| Cycle 7 audit patch | <code>tests/verify_dc005_merge_ready.py</code> | Narrowed DC-006 rejection to reuse of DC-005 TRJ-* IDs or DC-005 collapse thresholds. |
| Cycle 9 audit       | conductor merge decision                       | Confirms verifier, organization, and promise-ledger state are all green.              |